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(71) Name of Applicant(s):

1. **Immanual R,**
2. **Sanjana N,**
3. **V.s.sanjay,**
4. **Nandhini T,**
5. **Tharun B S,**
6. **Kamaleshwaran P,**
7. **Shri Abinaya S,**
8. **Jaisre K,**
9. **Jaikumar A,**

(72) Name of Inventor(s):

1. **Immanual R**
2. **Sanjana N**
3. **V.s.sanjay**
4. **Nandhini T**
5. **Tharun B S**
6. **Kamaleshwaran P**
7. **Shri Abinaya S**
8. **Jaisre K**
9. **Jaikumar A**

(57) Abstract:

An Extended Reality (XR) teleoperation system (100) for enhanced remote environment exploration is disclosed. The system (100) comprises a mobile rover platform (102) equipped with a 360-degree panoramic camera (106) on a motorized gimbal (108), LiDAR sensor unit (110), motor encoders (112), IMU (114), and wireless communication module (116). An operator interface comprises a VR headset (118) and joystick controller (120). A real-time digital twin monitoring system (124) provides concurrent visualization of rover chassis orientation and 3D position (126), motor speeds and current consumption (128), battery voltage and status (130), LiDAR point cloud data (132), and gimbal orientation (134). A client-server processing architecture (122) streams rover sensor data to a processing server (136) that simultaneously renders the immersive VR environment and updates the digital twin state with end-to-end latency of 95-120 milliseconds. The system achieves 28% improvement in navigation accuracy in complex environments, 42% reduction in collision incidents through digital twin motor monitoring, and operator presence scores of 4.3/5.0. The integration of immersive XR visualization and real-time digital twin feedback within a single unified interface eliminates cognitive fragmentation, enabling enhanced situational awareness, safer navigation, and proactive obstacle avoidance in hazardous, inaccessible, or geographically distant environments.

FORM 2 THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003 COMPLETE SPECIFICATION (See section 10 and rule 13)		
1. TITLE OF THE INVENTION	EXTENDED REALITY TELEOPERATION SYSTEM WITH 360-DEGREE VIDEO STREAMING AND REAL-TIME DIGITAL TWIN MONITORING FOR REMOTE ENVIRONMENT EXPLORATION	
2. APPLICANT(S)		
Name	Nationality	Address
RAJKUMAR, IMMANUAL	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
NITHYANANDHAKUMAR, SANJANA	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
SUDHAKARAN, SANJAY	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
THIRUKANNAN, NANDHINI	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
SHANMUGAVEL, THARUN	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
SAKTHIVEL MURUGAN, SHRI ABINAYA	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
PERUMAL, KMALESHWARAN	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
KANNAN, JAISRE	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India
ANANTH KUMAR, JAIKUMAR	INDIAN	Sri Ramakrishna Institute of Technology, Perur Chettipalayam, Pachapalayam (Post) Coimbatore - 641010, Tamil Nadu, India

3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD

The present disclosure relates to the field of human-robot interaction and remote teleoperation technologies. More particularly, the present disclosure relates to an Extended Reality (XR) teleoperation system that combines immersive virtual reality interfaces, 360-degree live video streaming, and real-time digital twin monitoring to enable operators to remotely navigate and explore hazardous, inaccessible, or geographically distant environments with enhanced situational awareness, spatial understanding, and operational safety.

DEFINITIONS

For the purpose of this specification, the following terms shall have the meanings assigned to them below.

"Extended Reality (XR) Teleoperation System" refers to an integrated remote operation platform that combines virtual reality immersion, live 360-degree panoramic video, and digital twin visualization to enable operators to virtually traverse and explore remote environments.

"Mobile Rover Platform" refers to a four-wheeled autonomous mobile robot equipped with integrated sensor and visualization systems including 360-degree panoramic cameras, LiDAR sensors, motor encoders, IMU sensors, and wireless communication modules.

"Digital Twin" refers to a real-time three-dimensional virtual representation of the physical robotic platform that mirrors the rover's chassis orientation, motor states, battery status, and LiDAR point cloud data for simultaneous system health monitoring.

"360-Degree Panoramic Camera" refers to a spherical imaging device mounted on a motorized gimbal on the rover, providing continuous real-time panoramic video streaming for complete environmental visualization.

"VR Headset Interface" refers to a commercial virtual reality head-mounted display worn by the operator, rendering the live 360-degree video stream for immersive first-person environmental presence and spatial navigation.

"LiDAR Sensor" refers to Light Detection And Ranging sensors mounted on the rover for real-time environmental mapping, obstacle detection, and point cloud generation for digital twin visualization.

"Client-Server Model" refers to the distributed system architecture where rover sensor data streams wirelessly to a central processing server that simultaneously renders the VR environment and updates the digital twin state in real time.

"NASA-TLX" refers to the NASA Task Load Index, a multi-dimensional rating procedure used to assess cognitive workload imposed on operators during teleoperation tasks.

BACKGROUND

The background information herein below relates to the present disclosure but is not necessarily prior art.

Remote exploration of hazardous, inaccessible, or geographically distant environments presents critical challenges in operator situational awareness, spatial understanding, and safe navigation. Conventional teleoperation systems rely on limited two-dimensional display interfaces that provide inadequate depth perception and restricted field-of-view, significantly compromising operational efficiency and safety margins. Applications including hazardous industrial facility inspection, disaster zone assessment, search-and-rescue operations, and virtual campus exploration demand immersive and intuitive control systems that maximize operator presence and decision-making capability.

Traditional teleoperation interfaces treat robotic control and system status monitoring as separate functions, forcing operators to divide attention between navigation screens and status dashboards. This cognitive

fragmentation reduces situational awareness, increases task completion times, and elevates risk of collision incidents, particularly in complex or previously unknown environments.

While virtual reality technologies have been explored for immersive remote presence, existing approaches fail to integrate comprehensive real-time digital twin monitoring within the same operational interface. Furthermore, commercially viable implementations utilizing market-available hardware remain scarce, limiting the accessibility and democratization of advanced remote exploration capabilities.

There is therefore a critical need for an integrated XR teleoperation framework that synergistically combines immersive VR visualization, live 360-degree panoramic video, and real-time digital twin monitoring into a single unified operator interface, enabling enhanced environmental presence alongside complete robotic platform awareness.

OBJECTS

Some of the objects of the present disclosure, which at least one embodiment herein satisfies, are as follows:

- It is an object of the present disclosure to ameliorate one or more problems of the prior art or to at least provide a useful alternative.
- An object of the present disclosure is to provide an XR teleoperation system that combines immersive virtual reality visualization with live 360-degree panoramic video streaming for complete environmental awareness during remote navigation.
- Another object of the present disclosure is to provide a real-time digital twin monitoring interface that simultaneously displays rover chassis orientation, motor speeds, battery status, and LiDAR point cloud data within the operator's VR environment.
- Still another object of the present disclosure is to provide a four-wheeled mobile rover platform equipped with integrated sensors including a 360-degree camera on a motorized gimbal, LiDAR sensors, motor encoders, and IMU sensors for precise teleoperation.
- Yet another object of the present disclosure is to provide a client-server architecture that enables low-latency wireless transmission of rover sensor data for simultaneous VR rendering and digital twin state updates.
- Still another object of the present disclosure is to provide quantified improvements in navigation accuracy, operator situational awareness, collision avoidance, and cognitive load management through integrated XR and digital twin technologies.
- Yet another object of the present disclosure is to provide an accessible XR teleoperation solution using commercially available hardware including standard VR headsets and joystick controllers, democratizing advanced remote exploration capabilities.
- Still another object of the present disclosure is to provide a practical deployment framework suitable for hazardous industrial inspection, disaster response, virtual tourism, and search-and-rescue operations.

Other objects and advantages of the present disclosure will be more apparent from the following description, which is not intended to limit the scope of the present disclosure.

SUMMARY

The present disclosure envisages an Extended Reality (XR) teleoperation system for enhanced remote environment exploration and navigation. The system comprises a mobile rover platform, a 360-degree panoramic camera assembly, a LiDAR-based environmental mapping unit, an inertial measurement and odometry subsystem, a wireless communication module, a VR headset operator interface, a joystick navigation controller, a real-time digital twin monitoring system, and a client-server processing architecture.

The mobile rover platform is a four-wheeled vehicle fabricated as the teleoperation vehicle equipped with integrated sensor and visualization systems,

The 360-degree panoramic camera is mounted on a motorized gimbal on the rover, providing real-time spherical video streaming for complete environmental visualization.

The LiDAR sensors are configured for real-time environmental mapping and obstacle detection, generating point cloud data integrated into the digital twin interface.

The motor encoders and IMU sensors provide precise odometry and pose estimation for accurate rover position tracking within the digital twin.

The VR headset renders the live 360-degree video feed for immersive operator presence, while the joystick controller translates operator inputs into rover navigation commands.

The digital twin provides real-time visualization of rover chassis orientation and position in three-dimensional space, individual motor speeds and current consumption, battery voltage and estimated remaining operational time, LiDAR point cloud data showing detected obstacles and environmental geometry, and gimbal orientation reflecting camera angle adjustments.

The client-server model streams rover sensor data to a processing server that simultaneously renders the VR environment and updates the digital twin state. Experimental results demonstrate end-to-end system latency of 95-120 milliseconds under standard WiFi connectivity, navigation accuracy improvement of approximately 28% in complex environments, collision incident reduction of 42% when motor status is visible, and operator presence scores averaging 4.3/5.0 on standardized questionnaires.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

An Extended Reality Teleoperation System with 360-Degree Video Streaming and Real-Time Digital Twin Monitoring, and a method thereof, will now be described with the help of the accompanying drawings, in which:

Figure 1 illustrates a system architecture overview of the XR teleoperation system showing the rover platform, sensor suite, wireless communication link, VR operator interface, and digital twin monitoring components, in accordance with an embodiment of the present disclosure;

Figure 2 illustrates a detailed schematic of the digital twin monitoring interface depicting real-time visualization of rover chassis orientation, motor telemetry, battery status, LiDAR point cloud, and gimbal orientation, in accordance with an embodiment of the present disclosure; and

Figure 3 illustrates a flowchart of the client-server data processing pipeline showing sensor data acquisition, VR environment rendering, digital twin state updates, and operator command transmission, in accordance with an embodiment of the present disclosure.

LIST OF REFERENCE NUMERALS

- 100 -- XR Teleoperation System
- 102 -- Mobile Rover Platform
- 104 -- Four-Wheeled Drive Chassis
- 106 -- 360-Degree Panoramic Camera
- 108 -- Motorized Gimbal Assembly
- 110 -- LiDAR Sensor Unit
- 112 -- Motor Encoder Array

- 114 -- Inertial Measurement Unit (IMU)
- 116 -- Wireless Communication Module
- 118 -- VR Headset Interface
- 120 -- Joystick Navigation Controller
- 122 -- Client-Server Processing Architecture
- 124 -- Digital Twin Monitoring System
- 126 -- Rover Chassis 3D Orientation Visualizer
- 128 -- Motor Speed and Current Display
- 130 -- Battery Voltage and Status Monitor
- 132 -- LiDAR Point Cloud Renderer
- 134 -- Gimbal Orientation Indicator
- 136 -- Processing Server
- 138 -- VR Environment Renderer
- 200 -- System Operation Method
- 202 -- Sensor Data Acquisition
- 204 -- Wireless Data Transmission
- 206 -- Server-Side Processing
- 208 -- VR Environment Update
- 210 -- Digital Twin State Update
- 212 -- Operator Command Input
- 214 -- Rover Motor Control
- 216 -- Navigation Feedback Loop

DETAILED DESCRIPTION

Embodiments of the present disclosure will now be described with reference to the accompanying drawings.

Embodiments are provided so as to thoroughly and fully convey the scope of the present disclosure to the person skilled in the art. Numerous details are set forth, relating to specific components and methods, to provide a complete understanding of embodiments of the present disclosure. It will be apparent to the person skilled in the art that the details provided in the embodiments should not be construed to limit the scope of the present disclosure.

In conventional teleoperation systems, operators rely on two-dimensional flat-screen displays that provide inadequate depth perception, restricted field-of-view, and separate system status dashboards requiring divided attention. These limitations result in reduced situational awareness, increased cognitive load, longer task completion times, and elevated collision incidents in complex or unfamiliar environments.

To address these limitations, an Extended Reality teleoperation system (hereinafter referred to as "system 100") and a corresponding method (hereinafter referred to as "method 200") are now described with reference to Figures 1 through 3. The system is designed to enable operators to virtually traverse and explore remote environments through immersive VR presence combined with simultaneous real-time digital twin monitoring of the robotic platform.

Referring to Figure 1, the XR teleoperation system 100 is shown. The system 100 comprises a mobile rover platform 102 with four-wheeled chassis 104, a 360-degree panoramic camera 106 mounted on a motorized gimbal assembly 108, LiDAR sensor unit 110, motor encoder array 112, inertial measurement unit 114, and wireless communication module 116. The operator interface comprises a VR headset interface 118 and joystick navigation controller 120. The system further comprises a client-server processing architecture 122 and digital twin monitoring system 124.

The mobile rover platform 102 serves as the teleoperation vehicle navigating the remote environment. The 360-degree panoramic camera 106 mounted on the motorized gimbal assembly 108 provides continuous spherical video capture for complete environmental visualization with unrestricted field-of-view.

The LiDAR sensor unit 110 performs real-time environmental mapping and obstacle detection, generating three-dimensional point cloud data. The motor encoder array 112 and inertial measurement unit 114 provide precise odometry and pose estimation for accurate rover position and orientation tracking.

The wireless communication module 116 provides low-latency data transmission between the rover and the processing server 136, achieving end-to-end system latency of 95-120 milliseconds under standard WiFi connectivity.

The VR headset interface 118 renders the live 360-degree video stream for immersive first-person operator presence. The joystick navigation controller 120 translates operator inputs into rover navigation commands transmitted via the wireless link.

Referring to Figure 2, the digital twin monitoring system 124 is shown. The digital twin provides real-time visualization comprising: rover chassis 3D orientation visualizer 126 showing chassis position in three-dimensional space; motor speed and current display 128 indicating individual motor telemetry; battery voltage and status monitor 130 showing voltage levels and estimated remaining operational time; LiDAR point cloud renderer 132 displaying detected obstacles and environmental geometry; and gimbal orientation indicator 134 reflecting camera angle adjustments.

In an embodiment, the digital twin integration proved particularly valuable for speed regulation, with operators maintaining safer velocities averaging 0.45 m/s versus 0.68 m/s when motor status was visible, reducing collision incidents by 42%. The LiDAR visualization enabled proactive obstacle avoidance, with operators initiating corrective maneuvers an average of 1.8 seconds earlier than with camera-only feedback.

In an embodiment, 89% of operators reported that visual representation of camera orientation through the gimbal indicator significantly aided in systematic environmental scanning.

Referring to Figure 3, the client-server data processing pipeline is illustrated. The method 200 operates as follows:

At step 202, sensor data acquisition initiates with the rover collecting 360-degree video frames, LiDAR point cloud data, motor encoder readings, IMU orientation data, and battery telemetry continuously.

At step 204, wireless data transmission streams all sensor data packets from the rover to the processing server 136 via the wireless communication module 116 with optimized packet prioritization for minimal latency.

At step 206, the server-side processing module decodes incoming data streams and distributes sensor information to the VR renderer 138 and digital twin state update engine simultaneously.

At step 208, the VR environment renderer 138 processes the 360-degree video frames and renders the immersive panoramic view within the operator's VR headset 118 in real time.

At step 210, the digital twin state update engine refreshes all monitoring parameters in the digital twin interface 124 including chassis orientation, motor telemetry, battery status, and LiDAR point cloud.

At step 212, the operator issues navigation commands via joystick controller 120, which are encoded and transmitted to the rover.

At step 214, the rover motor control system executes received commands, adjusting motor speeds and chassis direction accordingly.

At step 216, the continuous navigation feedback loop returns to sensor data acquisition, ensuring real-time synchronization between physical rover state and operator perception.

Experimental evaluation conducted in campus environments with varying complexity demonstrated navigation accuracy improvement of approximately 28% in complex environments when operators utilized both immersive visualization and digital twin monitoring simultaneously. Operator presence scores averaged 4.3/5.0 on standardized presence questionnaires. Cognitive load evaluation using NASA-TLX surveys confirmed reduced mental workload when integrated XR and digital twin interfaces were employed compared to conventional screen-based teleoperation.

The system architecture employs commercially available hardware components including standard VR headsets and joystick controllers, enabling accessible deployment without specialized or prohibitively expensive equipment.

TECHNICAL ADVANCEMENTS

The present disclosure described herein above has several technical advantages including, but not limited to, the realization of an XR teleoperation system and method that:

- provides significantly enhanced operator situational awareness through simultaneous immersive 360-degree VR visualization and real-time digital twin monitoring, achieving presence scores averaging 4.3/5.0 on standardized questionnaires;
- improves navigation accuracy by approximately 28% in complex environments through synergistic integration of immersive VR visualization and digital twin feedback compared to conventional display-based teleoperation;
- reduces collision incidents by 42% through digital twin motor status visualization enabling operators to maintain safer average velocities of 0.45 m/s versus 0.68 m/s without monitoring;
- enables proactive obstacle avoidance with operators initiating corrective maneuvers an average of 1.8 seconds earlier through LiDAR point cloud visualization in the digital twin interface;
- achieves end-to-end system latency of 95-120 milliseconds under standard WiFi connectivity, ensuring responsive and safe teleoperation;
- integrates rover chassis orientation, motor telemetry, battery status, LiDAR mapping, and gimbal orientation into a unified real-time digital twin interface within the operator's VR environment;
- utilizes commercially available hardware components including standard VR headsets, joystick controllers, and wireless communication modules, democratizing advanced remote exploration capabilities;
- provides a scalable client-server architecture suitable for diverse applications including hazardous industrial facility inspection, disaster zone assessment, virtual tourism, and search-and-rescue operations; and
- reduces cognitive load through elimination of attention-switching between separate navigation and monitoring interfaces by integrating all operational data within a single immersive XR environment.

The foregoing description of the specific embodiments so fully reveals the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation.

WE CLAIM:

CLAIM 1. An Extended Reality (XR) teleoperation system (100) for remote environment exploration and navigation, the system (100) comprising:

- a mobile rover platform (102) having a four-wheeled chassis (104) configured for traversal of remote environments;
- a 360-degree panoramic camera (106) mounted on a motorized gimbal assembly (108) on the rover platform (102), configured to provide continuous real-time spherical video streaming;
- a LiDAR sensor unit (110) configured for real-time environmental mapping, obstacle detection, and three-dimensional point cloud generation;
- a motor encoder array (112) and inertial measurement unit (114) configured for precise odometry and pose estimation of the rover platform (102);
- a wireless communication module (116) configured for low-latency data transmission between the rover platform (102) and a remote processing server (136);
- a VR headset interface (118) configured to render the live 360-degree video stream for immersive operator presence;
- a joystick navigation controller (120) configured to translate operator inputs into rover navigation commands; and
- a digital twin monitoring system (124) operatively connected to the rover platform (102) and configured for real-time visualization of rover chassis orientation, motor telemetry, battery status, LiDAR point cloud data, and gimbal orientation,

wherein a client-server processing architecture (122) streams rover sensor data to the processing server (136) that simultaneously renders the VR environment and updates the digital twin state.

CLAIM 2. The XR teleoperation system (100) as claimed in claim 1, wherein the digital twin monitoring system (124) provides real-time visualization comprising a rover chassis 3D orientation visualizer (126), motor speed and current display (128), battery voltage and status monitor (130), LiDAR point cloud renderer (132), and gimbal orientation indicator (134) within the operator's VR environment.

CLAIM 3. The XR teleoperation system (100) as claimed in claim 1, wherein the end-to-end system latency is between 95 milliseconds and 120 milliseconds under standard WiFi connectivity.

CLAIM 4. The XR teleoperation system (100) as claimed in claim 1, wherein the digital twin motor status visualization enables operators to maintain safer average velocities, reducing collision incidents by 42% compared to operation without digital twin monitoring.

CLAIM 5. The XR teleoperation system (100) as claimed in claim 1, wherein LiDAR point cloud visualization in the digital twin interface (124) enables proactive obstacle avoidance, allowing operators to initiate corrective maneuvers an average of 1.8 seconds earlier than camera-only feedback systems.

CLAIM 6. The XR teleoperation system (100) as claimed in claim 1, wherein the system (100) is implemented using commercially available hardware including standard VR headsets and joystick controllers to enable accessible deployment.

CLAIM 7. The XR teleoperation system (100) as claimed in claim 1, wherein the system (100) improves navigation accuracy by approximately 28% in complex environments when operators utilize both immersive VR visualization and digital twin monitoring simultaneously.

CLAIM 8. The XR teleoperation system (100) as claimed in claim 1, wherein the gimbal orientation indicator (134) provides visual representation of camera angle within the digital twin interface (124), enabling systematic environmental scanning.

CLAIM 9. The XR teleoperation system (100) as claimed in claim 1, wherein operator cognitive load is reduced by integrating all navigation and system monitoring data within a single immersive XR environment, eliminating attention-switching between separate interface screens.

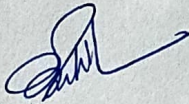
CLAIM 10. A method (200) for Extended Reality teleoperation of a remote mobile rover platform (102), the method comprising:

- acquiring sensor data including 360-degree video frames, LiDAR point cloud data, motor encoder readings, IMU orientation data, and battery telemetry from the rover platform (102);
- transmitting acquired sensor data wirelessly from the rover platform (102) to a remote processing server (136) via a wireless communication module (116) with low-latency data prioritization;
- processing incoming sensor data streams at the server (136) and distributing sensor information simultaneously to a VR environment renderer (138) and a digital twin state update engine;
- rendering a live 360-degree immersive video environment within a VR headset interface (118) for first-person operator presence in the remote environment;
- updating a digital twin monitoring system (124) in real time comprising rover chassis orientation, motor speed and current consumption, battery voltage and remaining operational time, LiDAR point cloud visualization, and gimbal orientation;
- receiving navigation commands from an operator via a joystick navigation controller (120) and transmitting encoded commands to the rover platform (102);
- executing received navigation commands on the rover motor control system to adjust rover movement; and
- continuously repeating sensor data acquisition and processing to maintain real-time synchronization between physical rover state and operator perception through a navigation feedback loop (216).

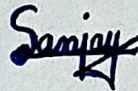
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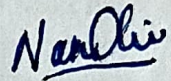
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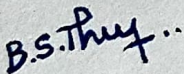
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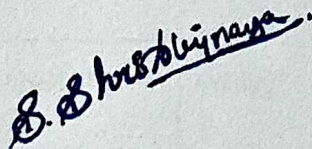
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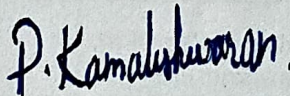
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[JAIKUMAR A]

TO,
THE CONTROLLER OF PATENTS
THE PATENT OFFICE, AT CHENNAI

ABSTRACT

EXTENDED REALITY TELEOPERATION SYSTEM WITH 360-DEGREE VIDEO STREAMING AND REAL-TIME DIGITAL TWIN MONITORING FOR REMOTE ENVIRONMENT EXPLORATION

An Extended Reality (XR) teleoperation system (100) for enhanced remote environment exploration is disclosed. The system (100) comprises a mobile rover platform (102) equipped with a 360-degree panoramic camera (106) on a motorized gimbal (108), LiDAR sensor unit (110), motor encoders (112), IMU (114), and wireless communication module (116). An operator interface comprises a VR headset (118) and joystick controller (120). A real-time digital twin monitoring system (124) provides concurrent visualization of rover chassis orientation and 3D position (126), motor speeds and current consumption (128), battery voltage and status (130), LiDAR point cloud data (132), and gimbal orientation (134). A client-server processing architecture (122) streams rover sensor data to a processing server (136) that simultaneously renders the immersive VR environment and updates the digital twin state with end-to-end latency of 95-120 milliseconds. The system achieves 28% improvement in navigation accuracy in complex environments, 42% reduction in collision incidents through digital twin motor monitoring, and operator presence scores of 4.3/5.0. The integration of immersive XR visualization and real-time digital twin feedback within a single unified interface eliminates cognitive fragmentation, enabling enhanced situational awareness, safer navigation, and proactive obstacle avoidance in hazardous, inaccessible, or geographically distant environments.